Fraction-of-50 Structure in CPTG and Related Cosmological Diagnostics

Research note with source citations

Prepared as a diagnostic catalogue for Curvature Polarization Transport Gravity (CPTG)

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Abstract. This note catalogues denominator-50 and nearby compact-rational structures in the CPTG galaxy-limit equations, the reduced transport framework, an exploratory early-expansion CMB transport-source run, selected cosmological quantities not specific to CPTG, and selected solar-system structural examples. Equation-level CPTG constants are separated from dimensionless diagnostics, rounded statistical outputs, and solar-system resonances or ratios. External cosmological and solar-system entries are included as comparative references only; they should not be read as evidence for CPTG unless they survive model-specific ablation, null testing, and independent reruns.

1 Scope and interpretation rules

A denominator-50 lattice is defined by

$$x_n = \frac{n}{50}, \qquad n \in \mathbb{Z}. \tag{1}$$

The catalogue records quantities that lie directly on this lattice, can be rewritten on it, or recur near nearby compact rational values. The evidentiary hierarchy is

$$\text{equation-level constants} > \text{dimensionless physical ratios} > \text{rounded fit statistics.} \tag{2}$$

The CPTG-specific equations are drawn from the uploaded CPTG manuscript, especially the reduced suppression law, galaxy-limit equation, baryonic source mapping, geometric transport remnant, structural mode definition, outer-regime attractor, and BTFR derivation [1]. CMB-source entries are drawn from the uploaded early-transport source-model summary [2]. External cosmological entries use Planck 2018 results, MOND literature, BTFR literature, and inflationary e-fold references [3,4,5,6,7,8]. Solar-system entries use NASA and JPL sources for orbital periods, resonances, distances, and Earth-Moon/Pluto-Charon comparison values [9,10,11,12,13,14].

2 Equation-level fraction structure in the CPTG paper

2.1 Reduced suppression law

The CPTG transport suppression factor is written as [1]

$$\Sigma(g) = \left[1 + \left(\frac{g}{a_\star}\right)^{123/50}\right]^{-11/50}. \tag{3}$$

The reduced galaxy-limit equation carries the same fractional suppression structure:

$$g(r) = g_{\rm bar}(r) + \frac{\sqrt{a_\star g_{\rm bar}(r)} + g_{\rm geom}(r)}{\left[1 + (g(r)/a_\star)^{123/50}\right]^{11/50}}. \tag{4}$$

The exponents

$$\frac{123}{50} = 2.46, \qquad \frac{11}{50} = 0.22 \tag{5}$$

are therefore equation-level response constants rather than visual pattern matches or output-format artifacts.

2.2 CPTG-weighted baryonic source field

In the reduced galaxy-limit source mapping, the gas, disk, and bulge components are weighted as [1]

$$g_{\rm bar}(r) = \frac{\frac{27}{50}v^2_{\rm gas}(r) + \frac{23}{50}v^2_{\rm disk}(r) + \frac{32}{50}v^2_{\rm bul}(r)}{r}. \tag{6}$$

The gas and disk coefficients form a closed complementary pair:

$$\frac{27}{50} + \frac{23}{50} = 1, \qquad \frac{1}{2}\left(\frac{27}{50} + \frac{23}{50}\right) = \frac{25}{50} = \frac{1}{2}. \tag{7}$$

The bulge coefficient is a compact-source weighting equal to 16/25, expressed above on the denominator-50 lattice as 32/50. It should be treated separately from the gas-disk closure pair.

2.3 Geometric transport remnant

The radial geometric transport remnant is [1]

$$g_{\rm geom}(r) = \eta a_\star \frac{(r/R_c)^{7/5}}{1+(r/R_c)^{12/5}}, \qquad \eta = \frac{1}{2}. \tag{8}$$

The powers and the averaging coefficient can be rewritten as

$$\frac{7}{5} = \frac{70}{50}, \qquad \frac{12}{5} = \frac{120}{50}, \qquad \eta = \frac{25}{50}. \tag{9}$$

These are equation-level radial-shape quantities. They are stronger evidence than post-fit numerical coincidences because they participate directly in the reduced field response.

2.4 CPTG scale relation and circular normalization

The CPTG acceleration scale is linked to the cosmological scale by [1]

$$a_\star = \frac{3}{16\pi} c H_0, \qquad H_0 = \frac{16\pi}{3}\frac{a_\star}{c}. \tag{10}$$

The coefficient

$$\frac{3}{16\pi} \simeq 0.05968 \simeq \frac{3}{50} \tag{11}$$

is not a denominator-50 constant in exact form, but it falls very close to a compact denominator-50 value because of the inverse circular normalization. This is best interpreted as a circular-normalization connection, not as an additional exact CPTG lattice coefficient.

3 Outer regime, structural mode, and BTFR scaling

3.1 Outer-regime attractor

The extended CPTG solution defines the local logarithmic rotation-curve slope as [1]

$$s(r) \equiv \frac{d\ln V}{d\ln r}, \qquad s_\infty = \frac{1}{4}. \tag{12}$$

Equivalently,

$$V(r) \propto r^{1/4}, \qquad g(r) \propto r^{-1/2}. \tag{13}$$

This is a quarter-power asymptotic attractor. It is not an integer $n/50$ entry, but it is part of the same compact-fraction response family and arises from the square-root low-acceleration polarization scaling.

3.2 Structural mode interval

The Curvature-Weighted Structural Mode Index is [1]

$$N \equiv \frac{R}{\lambda}. \tag{14}$$

With

$$w(r) = \left|\frac{dC}{dr}\right|, \tag{15}$$

the outer curvature-organization scale is

$$\lambda = \frac{\int_{R/5}^{R} r\, w(r)\, dr}{\int_{R/5}^{R} w(r)\, dr}, \qquad N = \frac{R}{\lambda}. \tag{16}$$

The lower radial cutoff $R/5$ can be written as $10/50$ of the resolved radius. This is a weak but explicit denominator-50-compatible interval in the structural-mode definition.

3.3 BTFR scaling

The baryonic Tully-Fisher relation is empirically expressed as a fourth-power relation between baryonic mass and characteristic rotation speed [7]. In CPTG, the leading low-acceleration scaling is [1]

$$g(r) \simeq \sqrt{a_\star g_{\rm bar}(r)}. \tag{17}$$

Using

$$g_{\rm bar}(r) \simeq \frac{GM_{\rm eff}}{r^2}, \qquad g(r) = \frac{V^2(r)}{r}, \tag{18}$$

the radial dependence cancels and yields

$$V_f^4 \simeq G a_\star M_{\rm eff}. \tag{19}$$

With $M_{\rm eff} = \Xi M_b$, this becomes

$$V_f^4 \simeq G a_\star \Xi M_b. \tag{20}$$

The fourth-power BTFR slope is therefore tied to the square-root polarization response and the CPTG source mapping, rather than imposed as a separate interpolation law.

4 CPTG CMB-source model diagnostics

4.1 Early-expansion clock values

The exploratory CMB transport-source summary records a Planck-like early-expansion clock with [2]

$$H_0 = 67.321170, \qquad \Omega_m = 0.31582310, \qquad \Omega_r = 9.27233359 \times 10^{-5}, \tag{21}$$

$$z_\star = 1089.914, \qquad z_{\rm eq} = 3405.080. \tag{22}$$

The equality-to-recombination ratios are

$$\frac{z_\star}{z_{\rm eq}} = \frac{1089.914}{3405.080} \simeq 0.32008, \tag{23}$$

$$\frac{1+z_\star}{1+z_{\rm eq}} \simeq 0.32028, \qquad \frac{1+z_{\rm eq}}{1+z_\star} \simeq 3.12223. \tag{24}$$

The inverse interval is close to π at the percent level. This makes the equality-to-recombination timing interval a useful diagnostic of compact inverse-circular scaling, independent of any claim that it is a CPTG constant.

4.2 CMB shell-geometry ratios

The best raw hard-shell source proxy and best cross-validated proxy give compact shell geometries [2]:

$$r_{\rm shell,raw} = \frac{178.370773}{552.181947} \simeq 0.3230, \tag{25}$$

$$r_{\rm shell,CV} = \frac{176.977272}{539.719208} \simeq 0.3279. \tag{26}$$

These ratios compare two fitted shell-scale quantities and therefore form dimensionless geometry. They are stronger diagnostics than rounded χ^2 values, but still require seed-repeat, block-holdout, and full-pipeline null tests.

4.3 CMB fit and validation scores

The same run reports positive TT, TE, and EE holdout improvements, with the best source proxy exceeding the calibrated 95% look-elsewhere null but not the 99% level [2]. The score values align closely with a denominator-50 reporting grid:

$$\Delta\chi^2_{\rm TT} \simeq 7.7774 \simeq \frac{389}{50}, \qquad \Delta\chi^2_{\rm TE} \simeq 7.3556 \simeq \frac{368}{50}, \tag{27}$$

$$\Delta\chi^2_{\rm EE} \simeq 2.0048 \simeq \frac{100}{50}, \qquad \sum \Delta\chi^2 \simeq 17.1377 \simeq \frac{857}{50}. \tag{28}$$

These entries are retained as weak diagnostics only. Statistical scores can land near simple grids through rounding and should not be weighted like equation-level constants or dimensionless shell geometry.

5 External cosmological connections not specific to CPTG

5.1 Planck 2018 parameter values

Planck 2018 reports, for base ΛCDM, representative values including $\Omega_c h^2 \simeq 0.120$, $\Omega_b h^2 \simeq 0.0224$, $n_s \simeq 0.965$, optical depth $\tau \simeq 0.054$, $100\theta_\star \simeq 1.0411$, $H_0 \simeq 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m \simeq 0.315$, $\sigma_8 \simeq 0.811$, and $N_{\rm eff} \simeq 2.99$ in the relevant combined analyses [3]. Several of these lie near denominator-50 grid values:

$$\Omega_c h^2 \simeq 0.120 \simeq \frac{6}{50}, \qquad \Omega_b h^2 + \Omega_c h^2 \simeq 0.1425 \simeq \frac{7}{50}, \tag{29}$$

$$\Omega_m \simeq 0.315 \simeq 0.32, \qquad \Omega_\Lambda \simeq 0.685 \simeq \frac{34}{50}, \tag{30}$$

$$\frac{\Omega_b h^2}{\Omega_b h^2 + \Omega_c h^2} \simeq 0.157 \simeq \frac{8}{50}, \qquad \frac{\Omega_c h^2}{\Omega_b h^2 + \Omega_c h^2} \simeq 0.843 \simeq \frac{42}{50}. \tag{31}$$

Additional weaker grid-near entries include

$$n_s \simeq 0.965 \simeq \frac{48}{50}, \qquad 100\theta_\star \simeq 1.0411 \simeq \frac{52}{50}, \tag{32}$$

$$\sigma_8 \simeq 0.811 \simeq \frac{41}{50}, \qquad \tau \simeq 0.054 \simeq \frac{3}{50}, \qquad N_{\rm eff} \simeq 2.99 \simeq \frac{150}{50}. \tag{33}$$

These external entries should be treated cautiously. Some are physical density parameters; others depend on conventional scaling choices, such as the factor of 100 in $100\theta_\star$.

5.2 Matter-radiation composition at recombination

Using the equality-to-recombination scale-factor ratio in the CMB proxy, the matter-to-radiation ratio at recombination is

$$\left.\frac{\rho_m}{\rho_r}\right|_{z_\star} = \frac{1+z_{\rm eq}}{1+z_\star} \simeq 3.12223 \simeq \frac{156}{50} \simeq \pi. \tag{34}$$

The corresponding matter and radiation fractions within the matter-plus-radiation budget are

$$f_m \simeq 0.7574 \simeq \frac{38}{50}, \qquad f_r \simeq 0.2426 \simeq \frac{12}{50}. \tag{35}$$

This is an external cosmological diagnostic derived from equality and recombination timing. It is not a CPTG equation-level constant, but it is relevant because CPTG's exploratory CMB program is explicitly focused on early-expansion transport before recombination.

5.3 MOND acceleration scale

MOND introduces a characteristic acceleration scale, conventionally quoted near [5,6]

$$a_0 \simeq 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{36}$$

The dimensionless coefficient is

$$1.2 = \frac{60}{50}. \tag{37}$$

This does not imply that MOND is a denominator-50 theory. It simply places the standard MOND acceleration coefficient on the same numerical reporting lattice used in this diagnostic catalogue.

5.4 Inflationary e-fold context

Inflationary cosmology often discusses observable perturbations in terms of roughly 50 to 60 e-folds before the end of inflation, with the exact number depending on assumptions about reheating and post-inflationary expansion [8,4]. This is not a denominator-50 fraction, but it is a direct recurrence of the number 50 in early-universe scaling arguments:

$$N_{\rm obs} \sim 50\text{–}60. \tag{38}$$

This entry is included for completeness only and should be classified as contextual, not as evidence for CPTG fraction structure.

6 Solar-system fraction-of-50 examples

Solar-system entries are included as external structural diagnostics. The strongest entries are dynamically maintained resonances, especially mean-motion resonances. Rounded AU distances and rounded catalog ratios are weaker because they depend on reporting precision. The purpose of this table is to retain the examples for watch-list comparison, not to treat the solar system as evidence for CPTG.

Table 1: Solar-system examples on or near the denominator-50 grid. Strength labels distinguish dynamical resonances from rounded distance and size ratios.

Structure or quantity	Source value	Fraction-of-50 form	Diagnostic interpretation	Source
Pluto-Neptune resonance	$3:2=1.5$	75/50	Strong. Pluto circles the Sun twice for every three Neptune orbits; this is a real mean-motion resonance, not a rounded ratio.	[11]
Jupiter-Saturn near resonance	$P_S/P_J \simeq 2.482$	near $5/2 = 125/50$; observed nearest 124/50	Strong-to-medium. Classical near $5:2$ period structure; observed ratio is close but not exactly locked.	[9]
Kirkwood gap, main belt	$3:1=3.0$	150/50	Strong. JPL identifies this as a primary Jupiter mean-motion resonance producing a Kirkwood gap.	[10]
Kirkwood gap, main belt	$5:2=2.5$	125/50	Strong. Primary Jupiter mean-motion resonance in the asteroid belt.	[10]
Kirkwood gap, main belt	$7:3\simeq 2.333$	near 117/50	Strong but not exactly integer-$n/50$. Included because JPL lists $7:3$ as a primary gap resonance.	[10]
Kirkwood gap, main belt	$2:1=2.0$	100/50	Strong. Primary Jupiter mean-motion resonance in the asteroid belt.	[10]
Charon/Pluto size	about 1/2	25/50	Medium. NASA describes Charon as about half Pluto's size, making Pluto-Charon double-planet-like.	[13]
Mercury/Earth orbital period	0.24084	near $12/50 = 0.24$	Medium. JPL sidereal periods place Mercury's year very close to a denominator-50 step.	[9]
Venus/Earth orbital period	0.61519	near $31/50 = 0.62$	Medium. Also overlaps the golden-ratio watch because $31/50 \approx 1/\phi$.	[9]
Earth/Earth orbital period	1	50/50	Weak. Included only as the normalization reference.	[9]
Mars/Earth orbital period	1.88081	near $94/50 = 1.88$	Medium. Very close to a denominator-50 step.	[9]
Jupiter/Saturn orbital-period ratio	0.40284	near $20/50 = 0.40$	Medium. Inverse view of the Jupiter-Saturn near-resonance.	[9]
Uranus/Neptune orbital-period ratio	0.50984	near $25/50 = 0.50$	Medium. Outer-planet period ratio near one-half.	[9]
Mercury distance from Sun	0.39 AU	near $20/50 = 0.40$	Weak. Rounded AU scale-model value; useful only as a watch-list entry.	[14]
Venus distance from Sun	0.72 AU	36/50	Weak. Rounded AU scale-model value.	[14]
Earth distance from Sun	1.00 AU	50/50	Weak. Definition of the AU normalization.	[14]
Mars distance from Sun	1.52 AU	76/50	Weak. Rounded AU scale-model value.	[14]

Structure or quantity	Source value	Fraction-of-50 form	Diagnostic interpretation	Source
Jupiter distance from Sun	5.20 AU	260/50	Weak. Rounded AU scale-model value.	[14]
Saturn distance from Sun	9.54 AU	477/50	Weak. Rounded AU scale-model value.	[14]
Uranus distance from Sun	19.20 AU	960/50	Weak. Rounded AU scale-model value.	[14]
Neptune distance from Sun	30.06 AU	1503/50	Weak. Rounded AU scale-model value.	[14]
Mercury/Venus distance ratio	$0.39/0.72 \simeq 0.5417$	near $27/50 = 0.54$	Weak-to-medium. Repeats a 27/50-class adjacent-planet ratio, but uses rounded AU inputs.	[14]
Jupiter/Saturn distance ratio	$5.20/9.54 \simeq 0.5451$	near $27/50 = 0.54$	Weak-to-medium. A second 27/50-class adjacent-planet distance ratio from rounded AU values.	[14]
Moon/Earth volume ratio	0.02028	near $1/50 = 0.02$	Medium. Literal 1/50-class Earth-Moon volume ratio using NASA listed volumes.	[12]
Moon/Earth surface-gravity ratio	0.16560	near $8/50 = 0.16$	Medium. Physical surface-gravity ratio; close to an inverse-circular 8/50 scale.	[12]
Moon/Earth equatorial-radius ratio	0.27272	near $14/50 = 0.28$	Medium. Size-ratio diagnostic; less direct than the volume ratio.	[12]
Moon/Earth escape-velocity ratio	0.21229	near $11/50 = 0.22$	Medium. Escape-velocity ratio from NASA listed values.	[12]
Moon orbital eccentricity	0.0554	near $3/50 = 0.06$	Weak-to-medium. A single orbital-shape parameter near a denominator-50 step.	[12]
Pluto perihelion/aphelion ratio	$30/49.3 \simeq 0.6085$	near $30/50 = 0.60$	Weak-to-medium. Uses NASA's Pluto perihelion and aphelion AU values.	[13]
Pluto average/aphelion ratio	$39/49.3 \simeq 0.7911$	near $40/50 = 0.80$	Weak-to-medium. Uses NASA's average Pluto distance and aphelion AU values.	[13]
Pluto-Charon double-body scale	Charon about half Pluto's size	25/50	Medium. Same physical example as Charon/Pluto size, retained as an outer-system structural entry.	[13]

The solar-system table suggests three interpretation tiers. First, resonance entries are the most meaningful because mean-motion resonances are real dynamical structures. Second, dimensionless ratios such as Moon/Earth volume or outer-planet period ratios are potentially useful diagnostics but do not imply CPTG. Third, rounded AU distances are retained only as weak grid-watch examples.

The solar-system entries also give a useful guardrail for the fraction-of-50 program: compact rational structure occurs naturally in gravitational systems through ordinary resonance and orbital dynamics. Therefore, fraction-of-50 recurrence is not by itself distinctive. What matters is whether a given fraction survives a theory-specific ablation, null, and cross-validation program.

7 Pi and circular normalization

The denominator-50 lattice approximates common circular constants at two-decimal precision:

$$\pi \simeq \frac{157}{50} = 3.14, \qquad 2\pi \simeq \frac{314}{50} = 6.28. \tag{39}$$

The inverse circular scale

$$\frac{1}{2\pi} \simeq 0.15915 \simeq \frac{8}{50} \tag{40}$$

is especially useful as a comparison point because baryon fractions and circular-normalization coefficients can land near it. The CPTG scale relation also contains $3/(16\pi)$, which falls near 3/50.

8 Diagnostic catalogue

Table 2: CPTG, cosmology-side, and solar-system fraction diagnostics. Strength labels distinguish equation-level constants from dimensionless ratios, rounded outputs, and external dynamical examples.

Quantity	Value	Grid form	Status and source
Suppression exponent	123/50	123/50	Equation-level CPTG response exponent [1].
Suppression power	11/50	11/50	Equation-level CPTG response exponent [1].
Gas source weight	27/50	27/50	Equation-level baryonic source coefficient [1].
Disk source weight	23/50	23/50	Equation-level baryonic source coefficient [1].
Bulge source weight	32/50	32/50	Equation-level compact-source coefficient [1].
Gas-disk closure	$27/50 + 23/50$	50/50	Complementary pair; high diagnostic value.
Gas-disk center	$(27/50 + 23/50)/2$	25/50	Central local-source response scale.
Transport exponent	7/5	70/50	Equation-level geometric remnant power [1].
Transport denominator exponent	12/5	120/50	Equation-level geometric remnant power [1].
Transport coefficient	1/2	25/50	Equation-level transport remnant coefficient [1].
Scale coefficient	$3/(16\pi)$	near 3/50	Circular-normalized CPTG scale coefficient [1].
Structural-mode cutoff	$R/5$	10/50 of R	Explicit interval in structural-mode definition [1].
Outer attractor	1/4	quarter-power	Compact asymptotic slope; not integer $n/50$ [1].
CDM physical density	0.120	6/50	External Planck-like parameter [3].
Matter physical density	0.1425	near 7/50	External Planck-like derived value [3].
Matter density fraction	0.315–0.316	near 0.32	External cosmological ratio [3].
Dark-energy fraction	0.684–0.685	near 34/50	External complementary ratio [3].
Baryon matter fraction	0.157	near 8/50	External matter-composition ratio [3].
CDM matter fraction	0.843	near 42/50	External matter-composition ratio [3].
Recombination/equality ratio	0.3203	near 0.32	CMB timing ratio from proxy values [2].
Equality/recombination expansion	3.1222	near 156/50	Close to π; external timing diagnostic.
Matter fraction at recombination	0.7574	near 38/50	Derived composition at recombination.
Radiation fraction at recombination	0.2426	near 12/50	Complementary derived composition.
Raw CMB shell ratio	0.3230	0.32-class	Dimensionless shell geometry [2].
CV CMB shell ratio	0.3279	0.32-class to 1/3	Cross-validated shell geometry [2].
Raw transport fraction	0.5532	near 28/50	Model diagnostic; medium strength [2].
CV transport fraction	0.5258	near 26/50	Model diagnostic; medium strength [2].
TT holdout score	7.7774	near 389/50	Weak; rounded statistical output [2].
TE holdout score	7.3556	near 368/50	Weak; rounded statistical output [2].
EE holdout score	2.0048	near 100/50	Weak; rounded statistical output [2].

Quantity	Value	Grid form	Status and source
Holdout sum	17.1377	near 857/50	Weak; rounded statistical output [2].
MOND coefficient	1.2	60/50	External modified-gravity scale coefficient [5,6].
Pluto-Neptune resonance	3 : 2	75/50	External solar-system mean-motion resonance [11].
Jupiter-Saturn near resonance	$P_S/P_J \simeq 2.482$	near 125/50	External solar-system near-resonance from JPL periods [9].
Kirkwood gaps	$3:1, 5:2, 7:3, 2:1$	150/50, 125/50, near 117/50, 100/50	External solar-system resonance family [10].
Charon/Pluto size	about 1/2	25/50	External outer-system size ratio [13].
Venus/Earth orbital period	0.6152	near 31/50	External planet-period ratio; phi-watch overlap [9].
Moon/Earth volume	0.02028	near 1/50	External Earth-Moon structural ratio [12].
Moon/Earth surface gravity	0.1656	near 8/50	External Earth-Moon structural ratio [12].
Mercury/Venus distance ratio	0.5417	near 27/50	External adjacent-planet ratio from rounded AU values [14].
Jupiter/Saturn distance ratio	0.5451	near 27/50	External adjacent-planet ratio from rounded AU values [14].
π	3.14159	near 157/50	Circular normalization benchmark.
2π	6.28318	near 314/50	Circular normalization benchmark.
$1/(2\pi)$	0.15915	near 8/50	Inverse circular normalization.
Inflation e-fold context	$N \sim 50$–60	direct 50-scale	Contextual early-universe recurrence [8,4].

9 Ablation and validation program

The galaxy-limit constants should be protected during exploratory development. If a one-step perturbation of an equation-level denominator-50 source or response constant degrades the benchmark sharply, the appropriate research action is to document the sensitivity as an ablation result rather than to tune new constants toward the lattice.

Recommended galaxy-limit ablations:

1. Perturb each equation-level coefficient by one lattice step and report the benchmark change.
2. Test the gas-disk pair with closure preserved and closure broken.
3. Test the compact bulge coefficient independently from the gas-disk pair.
4. Recompute SPARC benchmark margins against MOND and MOND-plus-Υ for every ablation.
5. Repeat with random seeds, larger galaxy subsets, and independent distance-quality cuts.

Recommended CMB and cosmology-side tests:

1. Keep denominator-50 tracking diagnostic-only in all interim source-proxy models.
2. Report nearest-lattice distance for dimensionless structural ratios, not only for raw χ^2 scores.
3. Add block holdouts in low-ℓ, shell-region, and high-ℓ intervals.
4. Add full-pipeline null tests, not only sign-flip nulls on stored templates.
5. Compare free, clock-tied, and constrained shell models only in the final revision stage.

Recommended solar-system-side handling:

1. Treat resonances as real dynamical structures, but not as CPTG evidence.
2. Treat rounded AU entries as weak watch-list examples only.
3. Track whether 27/50, 31/50, 25/50, and 1/50 recur in independent dimensionless ratios.
4. Keep phi and Fibonacci watches diagnostic-only. For example, $31/50 \approx 1/\phi$ and the $3:2$ resonance uses adjacent Fibonacci integers.

10 Research interpretation

The strongest denominator-50 structures are the constants embedded directly in the CPTG equations: the suppression exponents, the baryonic source weights, the radial transport-remnant powers, and the transport coefficient. The strongest extra-equation diagnostics are dimensionless ratios such as CMB shell width divided by shell center and early-expansion timing ratios. Solar-system resonances are important as comparative gravitational structure, but they are external examples with known ordinary dynamical explanations. Rounded fit statistics and rounded AU distances should remain secondary.

The correct development rule is

$$\text{observe and ablate first; constrain only in the final model revision.} \tag{41}$$

This rule permits denominator-50 structure to remain scientifically useful without becoming a self-fulfilling scoring target.

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